Euler Line — Three Centres, One Ratio

NOTEBOOK FLOW: read straight down the page, then continue on the next sheet.

LESSON 01 / 03

The vector identities

For a non-equilateral triangle with circumcentre O, centroid G and orthocentre H, use vectors measured from any origin. Since $H = A + B + C - 2O$ and $G = (A + B + C)/3$,

$$G - O = \tfrac{1}{3}(H - O).$$

Thus O, G, H are collinear and $OG : GH = 1 : 2$ (unless the triangle is equilateral, where $O = G = H$ and the line has no unique direction).

SHORT DERIVATION

Why the orthocentre vector works

Translate the origin to O, so $|A| = |B| = |C| = R$ and $H = A + B + C$. Then $H - A = B + C$. Its dot product with $C - B$ is $|C|^2 - |B|^2 = 0$, so $AH \perp BC$; cyclically, all three altitudes pass through H.

Add the nine-point centre

The nine-point centre is $N = (O + H)/2$. Along the Euler line the parameter positions from O to H are $0, 1/3, 1/2, 1$, so

$$O, G, N, H \quad \text{and} \quad OG : GN : NH = 2 : 1 : 3.$$

This is a length ratio, not the ratio of the four position coordinates.

WORKED EXAMPLE

A 3–4–5 right triangle

For $A = (0, 0), B = (4, 0), C = (0, 3)$, $O = (2, 3/2)$ and $H = A$. Then $G = (4/3, 1)$, $N = (1, 3/4)$, and the ordered distances along OH are in ratio $2 : 1 : 3$.

COMMON TRAP

Equilateral exception

In an equilateral triangle, all four classical centres coincide. Any statement that the Euler line has a direction or that the centres are distinct needs a non-equilateral hypothesis.

LESSON 02 / 03

The nine points

For triangle ABC, the side midpoints, the three altitude feet, and the midpoints of AH, BH, CH lie on one circle. Its centre is $N = (O + H)/2$ and its radius is $R/2$, where O, H and R are the circumcentre, orthocentre and circumradius.

SHORT DERIVATION

A coordinate proof for three of the nine points

Put the midpoint of BC at the origin and take BC as the x-axis. Write $B = (-s, 0)$, $C = (s, 0)$, circumcentre $O = (0, q)$, and $A = (u, v)$. Then $R^2 = s^2 + q^2 = u^2 + (v - q)^2$. The orthocentre vector identity gives $H = (u, v - 2q)$ and hence $N = (O + H)/2 = (u/2, (v - q)/2)$. The midpoint of BC is $(0, 0)$, the foot from A to BC is $(u, 0)$, and the midpoint of AH is $(u, v - q)$. Each differs from N by a vector with squared length $[u^2 + (v - q)^2]/4 = R^2/4$. Cycling the labels proves the same for the other six points.

WORKED EXAMPLE

Check all nine points in a non-right triangle

For $A = (0, 0), B = (4, 0), C = (1, 3)$, $O = (2, 1)$, $H = (1, 1)$, so $N = (3/2, 1)$ and $R = \sqrt{5}$. The three side midpoints are $(2, 0), (5/2, 3/2), (1/2, 3/2)$; the altitude feet are $(2, 2), (2/5, 6/5), (1, 0)$; the midpoints of AH, BH, CH are $(1/2, 1/2), (5/2, 1/2), (1, 2)$. Each is at distance $\sqrt{5}/2$ from N.

COMMON TRAP

A midpoint check alone is not the nine-point proof

The side-midpoint calculation does not automatically cover the altitude feet. The coordinate derivation above checks the side-midpoint, one altitude-foot and one midpoint-of-altitude cases, then cycles the labels. A few numerical examples alone would not establish the theorem for every triangle.

LESSON 03 / 03

Two distance constraints: translate before expanding

A condition involving distances to fixed points is often quadratic after squaring. Translate the midpoint of the fixed segment to the origin and rotate the axes to put the segment on the x-axis; translation and rotation preserve distances, so the algebra becomes one-dimensional in the fixed-point coordinates.

WORKED EXAMPLE

A ratio locus becomes a circle

With foci $A = (-c, 0)$ and $B = (c, 0)$, $PA/PB = k \neq 1$ yields

$$(x + c)^2 + y^2 = k^2((x - c)^2 + y^2).$$

Completing squares gives a circle on the x-axis. If $k = 1$, the quadratic terms cancel and the locus is $x = 0$, the perpendicular bisector. This is the Apollonius circle / bisector dichotomy.

A linear condition plus a distance condition

The intersection of a line and a circle has zero, one (tangent), or two points. Substituting the line parametrization into the circle gives a quadratic; its discriminant records these cases. The shortest-distance-to-line bound can often decide existence before solving.

COMMON TRAP

Locus proofs need both directions

An equation obtained by squaring is only a candidate locus until every point on it is checked in the original unsquared relation. Keep sign restrictions and any excluded denominator values.

ASSUMPTIONS & CONVENTIONS

Extension boundary

The coordinate methods here are extensions and problem-solving tools. The official JEE Advanced line syllabus lists the specific line forms, intersection-pencil, bisector, concurrency, and four triangle centres; it does not name every possible distance locus or every olympiad theorem.